WORKED SOLUTION

STEP II 2019, Question 6

Pure • First-Order Differential Equations

STEP II 2019, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

In both equations the right-hand side depends on x and y only through one combination: $y+x$ in part (i), and $y-x$ in part (ii), since the quadratic is really $(y-x)^2+4(y-x)+3$. That one observation finds the straight-line solutions, pins the stationary points to lines, and (in part (i)) turns the equation into one we can separate. The note about solution curves never crossing then does the rest: a curve that starts between two straight-line solutions is trapped there for ever.

Solution

Part (i)

The straight-line solution. If $y=mx+c$ then $\frac{\mathrm{d}y}{\mathrm{d}x}=m$, so we need

$$m=mx+c+x+1=(m+1)x+(c+1) \quad \text{for all } x.$$

Comparing coefficients of x gives $m=-1$, and then the constants give $c+1=m=-1$, so $c=-2$. The solution is $y=-x-2$.

Where the stationary points are. At a stationary point of $y=y_3(x)$ we have $0=\frac{\mathrm{d}y}{\mathrm{d}x}=y+x+1$, so the point lies on the line $y=-x-1$.

Now suppose $k<-2$. The curve $y=y_3(x)$ and the line $y=-x-2$ are both solutions, and at $x=0$ the curve is below the line ($k<-2$). They never meet, so the curve stays below the line for every x: $y_3(x)<-x-2<-x-1$. Hence $y_3(x)+x+1<-1<0$ everywhere. The gradient is never zero (in fact always less than -1), so there are no stationary points.

Every stationary point is a minimum. Differentiating the equation,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=\frac{\mathrm{d}y}{\mathrm{d}x}+1.$$

At a stationary point the first derivative is 0, so the second derivative is $1>0$, and the

point is a local minimum.

Solving. With $Y = y + x$ we have $\frac{\mathrm{d}Y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}x} + 1 = Y + 2$. On the line $Y = -2$ this is the solution we already have; otherwise we separate:

$$\int \frac{\mathrm{d}Y}{Y+2} = \int \mathrm{d}x \quad \Longrightarrow \quad \ln|Y+2| = x + \text{const} \quad \Longrightarrow \quad Y + 2 = Ae^x,$$

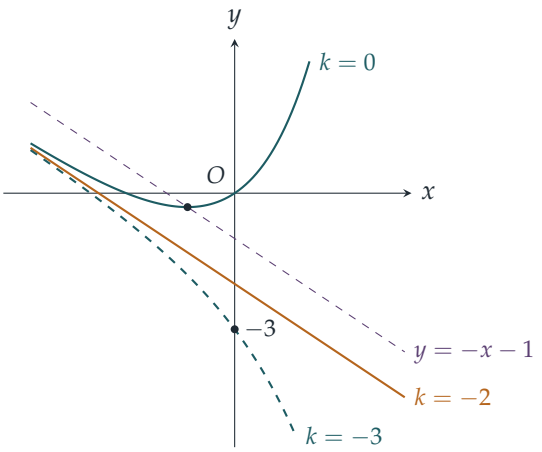
where A is a constant (and $A = 0$ covers the straight line). So $y = Ae^x - x - 2$, and $y(0) = k$ gives $A = k + 2$:

$$y_3(x) = (k+2)e^x - x - 2.$$

This agrees with what we found: $y_3'(x) = (k+2)e^x - 1$, which is never zero when $k \le -2$, and vanishes once, at $e^x = 1/(k+2)$, when $k > -2$.

The three curves.

- $k = 0$: $y = 2e^x - x - 2$, through the origin. The minimum is where $e^x = \frac{1}{2}$, at $\big(-\ln 2,\ \ln 2 - 1\big) \approx (-0.69, -0.31)$, on the line $y = -x - 1$. As $x \to -\infty$, $2e^x \to 0$ and the curve approaches $y = -x - 2$ from above; as $x \to \infty$ the exponential takes over.
- $k = -2$: the line $y = -x - 2$ itself.
- $k = -3$: $y = -e^x - x - 2$, through $(0, -3)$, always decreasing. It approaches $y = -x - 2$ from below as $x \to -\infty$ and plunges like $-e^x$ as $x \to \infty$.



Part (ii)

Rewriting the right-hand side. Grouping the terms,

$$x^2 + y^2 - 2xy - 4x + 4y + 3 = (y-x)^2 + 4(y-x) + 3 = (y - x + 1)(y - x + 3).$$

It is worth writing $u = y - x$, so that $\frac{\mathrm{d}y}{\mathrm{d}x} = (u+1)(u+3)$.

The straight-line solutions. If $y = mx + c$ then $u = (m-1)x + c$, and we need

$$m = \big((m-1)x + c + 1\big)\big((m-1)x + c + 3\big) \quad \text{for all } x.$$

If $m \neq 1$ the right-hand side is a genuine quadratic in x and can't equal a constant, so $m = 1$. Then $1 = (c+1)(c+3)$, that is $c^2 + 4c + 2 = 0$, giving $c = -2 \pm \sqrt{2}$. The two solutions are

$$y = x - 2 + \sqrt{2} \qquad \text{and} \qquad y = x - 2 - \sqrt{2}.$$

Stationary points of y_4. The gradient is zero exactly when $u = -1$ or $u = -3$, so any stationary point lies on

$$y = x - 1 \qquad \text{or} \qquad y = x - 3.$$

Between these lines we have $-3 < y - x < -1$, so $u + 1 < 0 < u + 3$ and the gradient $(u+1)(u+3)$ is *negative*. (Outside them, it is positive.)

Where the curve lives. At $x = 0$, $u = -2$, which lies between $-2 - \sqrt{2} \approx -3.41$ and $-2 + \sqrt{2} \approx -0.59$. So at $x = 0$ the curve is between the two straight-line solutions, and since solution curves never cross,

$$x - 2 - \sqrt{2} < y_4(x) < x - 2 + \sqrt{2} \quad \text{for all } x.$$

The band between the lines of stationary points ($-3 < u < -1$) sits inside this band.

Ask yourself The curve is trapped between two parallel lines, but where does it go inside the band?

Track $u = y_4 - x$. It satisfies

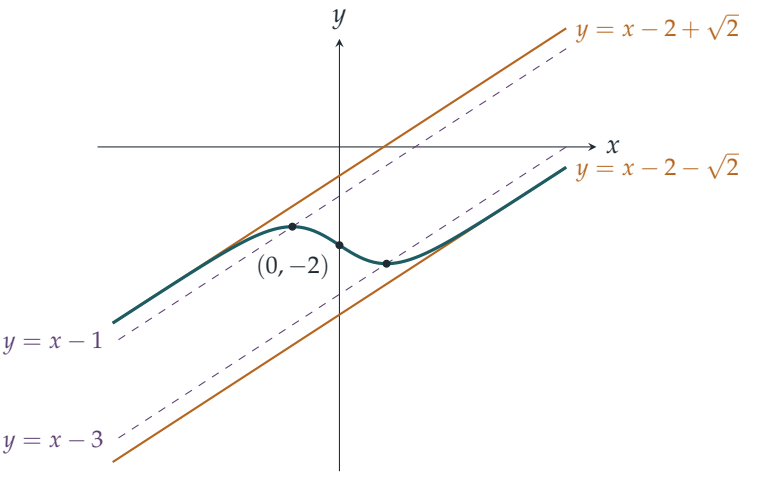
$$\frac{\mathrm{d}u}{\mathrm{d}x} = (u+1)(u+3) - 1 = u^2 + 4u + 2 = \left(u + 2 - \sqrt{2}\right)\left(u + 2 + \sqrt{2}\right),$$

which is negative while $-2 - \sqrt{2} < u < -2 + \sqrt{2}$, which is always. So u is strictly decreasing and bounded, and therefore has limits as $x \to \pm\infty$. At such a limit $\frac{\mathrm{d}u}{\mathrm{d}x}$ must tend to 0 (otherwise u would change without bound), so the limits are the roots: $u \to -2 + \sqrt{2}$ as $x \to -\infty$ and $u \to -2 - \sqrt{2}$ as $x \to +\infty$.

So, reading from left to right:

- far to the left the curve hugs $y = x - 2 + \sqrt{2}$ from below, with gradient close to 1;
- as u falls through -1 it meets $y = x - 1$ in a single point, a *local maximum* (the gradient changes from positive to negative);
- it falls through $(0, -2)$ with gradient $(-1)(1) = -1$;
- as u falls through -3 it meets $y = x - 3$ in a single point, a *local minimum*;
- then it rises again, hugging $y = x - 2 - \sqrt{2}$ from above with gradient tending to 1.

Each of $u = -1$ and $u = -3$ is passed exactly once because u is strictly decreasing, so there are exactly two stationary points. One more nice feature: replacing (x, y) by $(-x, -4 - y)$ leaves the equation and the starting point $(0, -2)$ unchanged, so the curve has half-turn symmetry about $(0, -2)$, and the maximum and minimum are mirror images of each other through that point.



Checks

- In part (i), substitute back: $y = (k+2)e^x - x - 2$ gives $y' = (k+2)e^x - 1$ and $y + x + 1 = (k+2)e^x - 1$. Setting $k = -2$ recovers the straight line.
- For $k = 0$ the minimum $(-\ln 2, \ln 2 - 1) \approx (-0.693, -0.307)$ does lie on $y = -x - 1$.
- In part (ii), a step-by-step numerical solution from $(0, -2)$ (confirmed by computer) stays between the two lines, reaches a maximum near $(-0.623, -1.623)$ on $y = x - 1$ and a minimum near $(0.623, -2.377)$ on $y = x - 3$, and by $x = \pm 6$ is within about 10^{-7} of the asymptotes. Those two points are images of each other in $(0, -2)$, as the symmetry predicts.

Common slips

- Forgetting to use the non-crossing property: "$k < -2$ means no stationary points" needs the curve to stay below $y = -x - 2$ for *all* x, not just at $x = 0$.
- Sketching y_4 crossing one of the straight-line solutions, or with the maximum and minimum the wrong way round. The sign of $(u+1)(u+3)$ between the dashed lines settles which is which.
- In finding m and c in part (ii), dividing by $m - 1$ without first ruling out (or in) $m = 1$.

Beyond STEP

The note at the start of the question is the uniqueness half of the Picard–Lindelöf theorem: for a smooth right-hand side, exactly one solution passes through each point, so solution curves cannot cross. Part (ii) is an *autonomous* equation in disguise, $u' = u^2 + 4u + 2$, and the way we found where the curve goes, from the equilibria and

the sign of u' in between and with no formula, is the start of the qualitative theory of differential equations. Boyce and DiPrima, *Elementary Differential Equations*, covers both uniqueness and this phase-line picture.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.